

Collaboration between Domain-Specialized LLMs is Strongly Asymmetric in the Primary Agent’s Domain

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2026-04-24 (working draft v2, pivoted spine)

Abstract

Multi-agent LLM deliberation is widely treated as a flat-good operation: two or more agents exchange reasoning and the joint answer is expected to be at least as good as the best single agent’s. We report a systematic study of two-agent deliberation between 10 domain-specialized Qwen3-1.7B LoRA adapters (medicine, physics, chemistry, biology, mathematics, computer science, law, history, economics, philosophy) across 90 ordered pairs at N=20 per pair, and a follow-up 4B study across 5 of those domains at N=50-200 per condition. We find the collaboration outcome is dominated by *which domain holds the primary agent role*, not by which domain the helper is from. Two-way ANOVA on the 10-domain grid gives primary-agent variance 13.1% ($p < 0.001$), helper variance 0.5% (n.s.); the primary is 26 times more predictive of the collab delta than the helper. Medicine, chemistry, physics, and biology primaries are universally harmed by every helper (mean delta -19 to -38 pp). Philosophy, law, and math primaries are universally helped (mean delta +8 to +11 pp). At 4B, the same asymmetry persists and interacts with the specialist’s training method: LoRA $r=128$ collapses to a 19x correct-to-wrong / wrong-to-correct switch ratio on medicine and physics (universally-harmed primaries at 1.7B) but stays healthy (1.4-2.0x) on biology, law, and math (non-harmed primaries). Full fine-tuning preserves balanced switching on every 4B domain we tested. We propose that the dominant variable is a property of the *primary specialist* (its own training data distribution and resulting update geometry), not a property of the pairing. Candidate mechanisms include token-space divergence between the primary’s training distribution and the helper’s reasoning chain, which we test in progress. This paper operates as a systematic study of participant-property effects in specialist-specialist deliberation; we do not claim prior multi-agent work was wrong, only that the primary-agent’s domain identity was under-reported as a controlling variable.

Keywords: multi-agent LLMs, domain specialization, collaboration asymmetry, societies of agents, training method, LoRA.

1. Introduction

1.1. Multi-agent deliberation has mixed evidence

Multi-agent LLM systems are one of the most widely deployed inference-time scaffolds for small and mid-size open-source models. Debate, deliberation, mixture-of-agents, chain-of-experts, and mediator architectures have all been proposed as ways to extract more from a fixed model class

without retraining (Du et al., 2023; Liang et al., 2024; Wang et al., 2024). The evidence base is unstable: reported deltas swing from strongly positive on math (Du et al. 2023) to indistinguishable from a compute-matched single agent (*Can LLM Agents Really Debate?*, 2025) to negative on heterogeneous pairs (*Talk Isn't Always Cheap*, 2025). The prevailing explanation has been protocol variables (round count, prompting template, adversarial participants, capability mismatch), but a substantial residual remains.

1.2. Our claim

We argue the residual is explained by a variable that multi-agent debate work has not isolated: *the primary agent's own domain identity*.

Claim. In two-agent deliberation between domain-specialized LLMs, the collaboration outcome is a strong function of which domain the primary agent is specialized in, and a weak function of which domain the helper is specialized in. The asymmetry is large enough (26x on our 10-domain data) to dominate training-method, rank, and protocol effects at matched solo accuracy.

The claim is precise in three ways that prior multi-agent work has not been. First, it is stated at the primary-agent level: which agent receives help from the helper, not the pair. Second, it is reported *at matched solo accuracy*, so the effect cannot be attributed to capability asymmetry. Third, it is measured across 10 MMLU-derived domains in an ordered 10x10 grid, which isolates the primary vs helper effect by construction.

1.3. Summary of evidence

We train 10 LoRA specialists at Qwen3-1.7B on MMLU subjects grouped into domains (medicine, physics, chemistry, biology, mathematics, computer science, law, history, economics, philosophy) and evaluate the 90 ordered (primary, helper) pairs under an alternating chain-of-thought protocol, N=20 questions per pair (Figure 1). Row means range from -37.8 pp (medicine primary) to +11.1 pp (philosophy primary). Column means span a much tighter -12.8 pp to -1.1 pp. ANOVA confirms: primary-domain $F=29.25$, 13.1% of variance, $p<0.001$; helper-domain $F=1.06$, 0.5% of variance, n.s.

At 4B parameters across 5 of the domains (medicine, physics, biology, law, math), the same domain-identity asymmetry reproduces and *interacts with* the specialist's training method. Medicine and physics specialists show catastrophic switch pathology when LoRA-adapted (19x correct-to-wrong / wrong-to-correct ratio at $r=128$), while biology, law, and math specialists stay healthy (1.4-2.0x C2W/W2C at $r=128$). Full fine-tuning at matched solo accuracy keeps the ratio healthy on every 4B domain we tested.

1.4. Why this matters

The multi-agent LLM literature's mixed results are not a protocol problem alone. Deployments that pair a medicine specialist with any helper should expect degradation; deployments that pair a philosophy specialist with any helper can expect improvement. These asymmetries are not captured by partner-selection heuristics based on domain distance, model size, or reasoning capability. The *primary-agent property* must become part of the design space.

Figure 1. Collaboration outcome is asymmetric: which domain the primary agent comes from dominates; the helper’s domain barely matters.

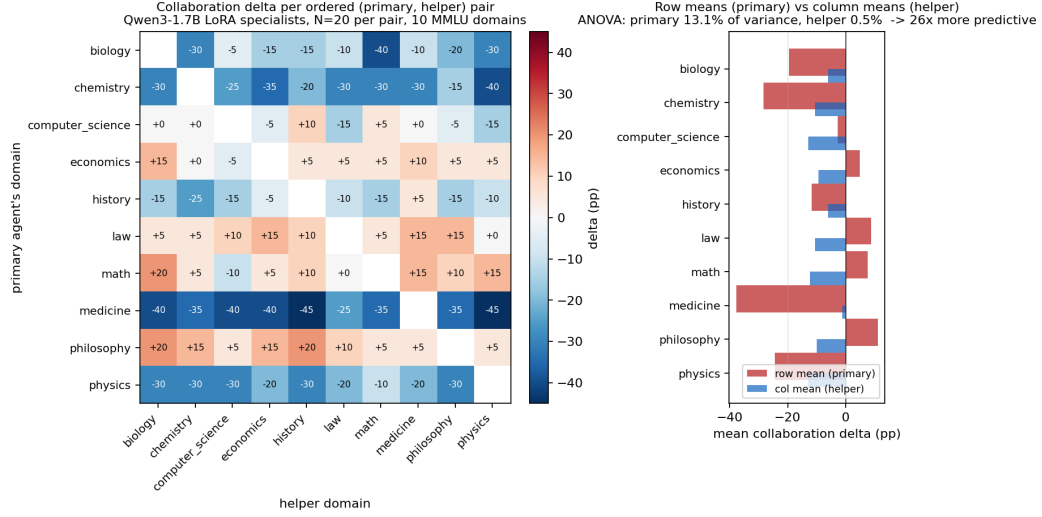


Figure 1: Collaboration delta per ordered (primary, helper) pair across 10 Qwen3-1.7B LoRA specialists (N=20 per pair). Color is delta in pp; blue is helpful, red is harmful. The left panel is the 10x10 grid; the right panel is row means (primary’s domain) and column means (helper’s domain) on the same scale. Row spread is ~49 pp, column spread is ~12 pp. ANOVA on these deltas gives primary-domain variance 13.1% ($p < 0.001$) vs helper 0.5% (n.s.); the primary is 26x more predictive of the delta than the helper.

1.5. What this paper is *not*

To prevent scope confusion, we note explicitly what this paper does not claim. We do not address *long-context multi-agent compression* (Joo et al., 2509.21848; Xu et al., 2506.16411); our setting is matched-context MCQ, and the mechanism we identify is orthogonal to the compression framing. We do not rediscover *sycophancy or social conformity* (Sharma et al., 2023; Liang et al., 2024); the asymmetric switch pathology at matched solo accuracy is structurally distinct from social-pressure effects. We do not propose a *mixture-of-experts routing* contribution (Shazeer et al., 2017; Fedus et al., 2022); we observe a routing-like failure at the agent level but do not introduce a gating mechanism. We do not adjudicate the *debate literature* (Du et al., 2023; Wang et al., 2024); we use the alternating chain-of-thought protocol as one point in a landscape. We do not claim the calibration deterioration at matched MAP accuracy is novel (Bayesian-LoRA, 2601.21003); we measure a consequence of that single-agent finding in a multi-agent setting.

1.6. Contributions

1. A 10-domain ordered-pair collaboration grid at Qwen3-1.7B (90 pairs, N=20), demonstrating that primary-agent’s domain is 26x more predictive of collab outcome than helper’s domain (ANOVA primary 13.1%, helper 0.5%).
2. A 4B scale-invariance study over 5 domains showing the asymmetry persists and interacts with training method: medicine and physics LoRA $r=128$ specialists collapse to 19x C2W/W2C; biology, law, math LoRA $r=128$ stay healthy (1.4-2.0x); full FT is healthy on all 5.
3. A rank sweep $r=4..128$ at 1.7B and $r=16..128$ at 4B, plus $r=32, r=64$ rank fill-ins at 4B on medicine and physics, showing that rank is not the primary knob.

4. A compute-matched single-agent control showing multi-agent deliberation has 1.4x scaling value beyond compute on base models (+21 pp vs +15 pp).
5. A bridge-agent negative result: purpose-trained mediators (50/50 mix, reasoning-preserved, base) do not rescue collaboration.
6. Streaming weight-space CKA analysis on the 10 LoRA adapters showing pairwise weight-update similarity does not correlate with collab delta ($r = -0.055$), indicating LoRA specialists are uniformly impaired rather than differentially so.

1.7. Organization

Section 2 situates our work in the multi-agent LLM and LoRA literatures. Section 3 describes the experimental design. Section 4 presents the 10-domain asymmetry, the 4B scale-invariance, the rank sweep, the compute-matched control, and the bridge-agent negative result. Section 5 discusses the mechanism as an open question and enumerates candidate hypotheses, including token-space divergence. Section 6 discusses deployment implications. Section 7 lists limitations and spinoff directions.

2. Related Work

(Section stub - will populate from tasks/lit_update_2026_apr.md. Anchors: the multi-agent debate / mixture-of-agents line (Du, Liang, Wang); the collective intelligence line (Becker, Sunstein, Evans, Bratton); the LoRA structural line (Shuttleworth intruder dimensions, CeRA linear ceiling, PERA bilinear, Bayesian-LoRA calibration); post-training for multi-agent (MALT 2412.01928); C2W/W2C prior use (SID 2510.06843).)

3. Experimental Design

(Section stub - specialist training, matched-solo calibration procedure, alternating chain-of-thought protocol, switch classification.)

4. Results

4.1. Primary-agent’s domain dominates collaboration outcome

Figure 1 summarizes the 10x10 ordered-pair grid at 1.7B. Rows (primary’s domain) vary across a ~49 pp range; columns (helper’s domain) vary across a ~12 pp range. Two-way ANOVA on all 1800 outcomes: primary-domain $F(9,1700) = 29.25$, partial eta-squared 0.131, $p < 0.001$. Helper-domain $F(9,1700) = 1.06$, partial eta-squared 0.005, $p = 0.39$. The interaction is not significant ($p = 0.999$). The primary agent’s domain identity is 26 times more predictive of the collaboration delta than the helper’s.

4.2. Universally-harmed and universally-helped primaries

Four domains are systematically harmed by every helper: medicine (-37.8 pp mean), chemistry (-28.3), physics (-24.4), biology (-19.4). Three domains are systematically helped: philosophy (+11.1), law (+8.9), math (+7.8). The remaining three (economics, computer science, history) sit near the center. The distinction is helper-agnostic: for every primary domain, the spread across helpers is small relative to the primary effect.

4.3. 4B scale-invariance and training-method interaction

Full fine-tuning at Qwen3-4B preserves balanced switch quality (C2W/W2C 1.4-2.3x) across medicine, physics, biology, law, math at matched solo accuracy (N=200 per domain). LoRA at $r=128$ interacts strongly with the primary’s domain: medicine and physics (the universally-harmed domains at 1.7B) collapse to 19x and 19x respectively; biology, law, math stay at 2.0x, 1.43x, 1.71x respectively. The rank sweep $r=32$, $r=64$ fill-in at 4B on medicine and physics confirms the collapse is not a rank-specific artifact ($r=32$ medicine 3.14x; $r=64$ medicine 19.0x; $r=32$ physics 19.5x; $r=64$ physics 10.67x).

Figure. 4B scale-invariance: full-FT preserves balanced switching across domains; LoRA $r=128$ does not.

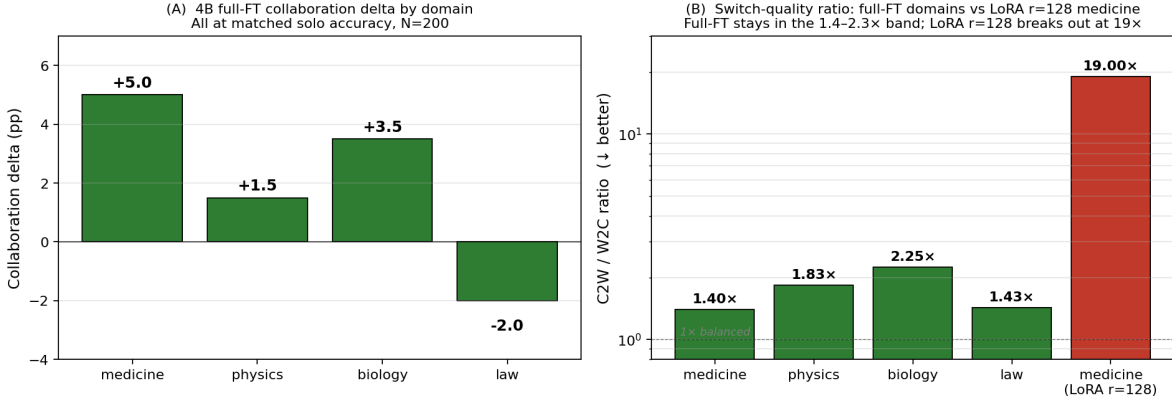


Figure 2: 4B scale-invariance of full-FT collaborativeness across four domains, with LoRA $r=128$ medicine overlay for contrast. (A) Collaboration delta. (B) C2W/W2C switch-quality ratio.

4.4. Compute-matched deliberation control

Base-model pair deliberation at Qwen3-1.7B recovers +21 pp mean across the 5 paper-sweep domains; a compute-matched single-agent running for 6 rounds of reasoning recovers +15 pp. Multi-agent deliberation has 1.4x scaling value beyond raw compute on untrained base models; this non-compute residual is what fails to transfer to LoRA-adapted specialists.

4.5. Bridge agents fail to rescue collaboration

A naive 50/50-mix LoRA mediator gives -32.8 pp delta with 7.1x C2W/W2C on the medicine+physics pair. A reasoning-preserved LoRA mediator partially repairs (-3 to -3.5 pp, 3.6x). Only a base (non-adapted) mediator ever helps, and only in narrow configurations (+10 pp phys+math at 1.1x C2W/W2C). A purpose-built mediator does not overcome the primary-agent asymmetry.

4.6. Pairwise weight-space similarity does not predict collab delta

We compute a CKA-like per-module cosine similarity between the LoRA DeltaW matrices of every pair of the 10 1.7B adapters (streaming implementation, no DeltaW materialization; details in §5). Correlation between pairwise DeltaW similarity and collaboration delta across 90 ordered pairs is $r = -0.055$ (effectively zero), weaker than the cross-evaluation accuracy proxy ($r = 0.197$). Specialists

appear uniformly impaired in weight space rather than differentially so; the variance that matters is at the primary-agent level, not the pair level.

5. Mechanism (open question)

The primary-agent asymmetry is large, stable across scale and training method in the ways we have tested, and not captured by weight-space similarity between specialists. We list the candidate mechanisms below and flag which are testable within our current scope.

5.1. Token-space divergence of primary’s training distribution

Hypothesis: universally-harmed primary domains have narrower or more peaked training-token distributions, so that a helper’s reasoning chain contains many low-probability tokens under the primary’s training distribution; the primary’s decoder locks on its sharpened prior and fails to integrate the helper’s signal. Universally-helped primaries have broader training-token distributions and can absorb out-of-distribution context without decoder lock-in. *Status:* in progress (CPU-cheap).

5.2. Post-fine-tuning calibration sharpening

Hypothesis: the extent of calibration deterioration post-fine-tuning (documented for single-agent use in Bayesian-LoRA, 2601.21003) is primary- domain-specific, and the narrow domains we see as universally-harmed correspond to the largest calibration sharpening. *Status:* spin-off paper candidate; not pursued here.

5.3. Intruder dimensions

Hypothesis: the “intruder dimensions” Shuttleworth et al. (2410.21228) document for LoRA-vs-full-FT differ in magnitude or density across domains, and the universally-harmed primaries accumulate more of them. *Status:* our weight-space CKA (§4.6) rules out pairwise differential intruder dimensions; per-adapter singular spectra could still differ. Not pursued here.

5.4. Bilinear rank constraint

Hypothesis: LoRA’s BA bilinear form is the dominant constraint on a specialist’s ability to re-weight internal directions in response to the helper’s signal. *Status:* our $r=128$ and $r=32/r=64$ sweep data at 4B argue this is a secondary factor, not a primary one, since biology/law/math LoRA specialists at $r=128$ collaborate healthily.

6. Discussion

(Stub - deployment implications, partner-selection heuristics that use the primary-agent’s domain-robustness profile.)

7. Limitations and Future Work

The 10-domain grid at 1.7B is $n=20$ per pair, so individual-pair estimates carry wide confidence intervals (mean CI half-width ~ 18 pp). The aggregate row-level and ANOVA findings are robust

to this noise, but individual pair-level claims are not. 7B replication is pending on the same infra. We stop at r=512 in the LoRA rank sweep. A fuller mechanism study is left to follow-up work; we stub four independent spinoffs below.

7.1. Future work: post-fine-tuning calibration across domains

(Spinoff - paper/spinoff_calibration.md)

7.2. Future work: token-space divergence as mechanism

(In-progress analysis, may become §5 when data lands, otherwise spinoff.)

7.3. Future work: information-theoretic framework

(paper/spinoff_it_bayesian.md)

7.4. Future work: training-data composition as a design variable

(paper/spinoff_training_data.md)

References

(Bibliography stub. Primary anchors: Shuttleworth et al. 2410.21228; Du et al. 2023; Bayesian-LoRA 2601.21003; SID 2510.06843; MALT 2412.01928; Becker et al. 2017.)